

2303A51200

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING	
Program Name: B. Tech		Assignment Type: Lab	Academic Year: 2025-2026
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CourseCode	23CS002PC304	Course Title	AI Assisted Coding
Year/Sem	III/II	Regulation	R23
Date and Day of Assignment	Week2 –	Time(s)	23CSBTB01 To 23CSBTB52
Duration	2 Hours	Applicable to Batches	All batches
Assignment Number: 4.1 (Present assignment number)/ 24 (Total number of assignments)			
Q.No.	Question		Expected Time to complete

1	Question 1: Zero-Shot Prompting (Leap Year Check) Write a zero-shot prompt to generate a Python function that checks whether a given year is a leap year. Task: Record the AI-generated code. Test with years like 1900, 2000, 2024. Identify logical flaws or missing conditions. Prompt Write a Python function that checks whether a given year is a leap year. CODE <pre>Copy > ... #Write a zero-shot prompt to generate a Python function that checks whether a given year is a leap year or not. def is_leap_year(year): """ Check if a given year is a leap year. A year is a leap year if it is divisible by 4, except for end-of-century years, which must be divisible by 400. Args: year (int): The year to check. Returns: bool: True if the year is a leap year, False otherwise. """ if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0): return True else: return False # Example usage: year = 2020 if is_leap_year(year): print(f"{year} is a leap year.") else: print(f"{year} is not a leap year.") year = 2020 if is_leap_year(year): print(f"{year} is a leap year.") else: print(f"{year} is not a leap year.")</pre> Code Output <pre>PS C:\Users\konda\Downloads\AI Assistant> & C:/Users/konda/AppData/Local/Microsoft/WindowsApps/python3.13.exe "d stant/day3.py" • 2020 is a leap year. 2020 is a leap year.</pre> Explanation • The function checks if the year is divisible by 4. • If yes, it returns True, otherwise False. Question 2: One-Shot Prompting (GCD of Two Numbers) Write a one-shot prompt with one example to generate a Python	Week2 -
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function that finds the Greatest Common Divisor (GCD) of two numbers.

Example:

Input: 12, 18 $\rightarrow$ Output: 6

Task:

Compare with a zero-shot solution.

Analyze algorithm efficiency.

Prompt

Write a Python function to find the GCD of two numbers.

Code

```
#Write a Python function to find the GCD of two numbers.
def gcd(a, b):
    while b:
        a, b = b, a % b
    return a

# Example usage:
num1 = 48
num2 = 18
print(f"The GCD of {num1} and {num2} is {gcd(num1, num2)}")

# Output: The GCD of 48 and 18 is 6

#Write a Python function to find the LCM of two numbers.
def lcm(a, b):
    return abs(a * b) // gcd(a, b)

# Example usage:
num1 = 4
num2 = 5
print(f"The LCM of {num1} and {num2} is {lcm(num1, num2)}")
```

Code Output

```
PS C:\Users\konda\Downloads\AI Assistant> & C:/Users/konda/AppData/Local/Programs/Python/Python39-64/Python.exe -i start/day3.py
The GCD of 48 and 18 is 6
The LCM of 4 and 5 is 20
```

Explanation

- Uses Euclidean Algorithm
- Repeatedly replaces (a, b) with $(b, a \% b)$
- Stops when $b = 0$
- Final value of a is the GCD

Question 3: Few-Shot Prompting (LCM Calculation)

Write a few-shot prompt with multiple examples to generate a Python function that computes the Least Common Multiple (LCM).

Examples:

Input: 4, 6 → Output: 12

Input: 5, 10 → Output: 10

Input: 7, 3 → Output: 21

Task:

Examine how examples guide formula selection.

Test edge cases.

Prompt

Write a Python function to calculate LCM of two numbers.

Code

```
#Write a Python function to calculate LCM of two numbers.
def calculate_lcm(x, y):
    if x > y:
        greater = x
    else:
        greater = y

    while True:
        if greater % x == 0 and greater % y == 0:
            lcm = greater
            break
        greater += 1

    return lcm

# Example usage
num1 = 12
num2 = 15
lcm = calculate_lcm(num1, num2)
print(f"The LCM of {num1} and {num2} is {lcm}")
```

Code Output

```
The LCM of 4 and 3 is 12
PS C:\Users\konda\Downloads\AI Assistant> & C:/Users/konda/AppData/Local/Microsoft
start/day3.py"
The LCM of 12 and 15 is 60
```

Explanation

- Uses the formula:

- $LCM(a, b) = (a \times b) / GCD(a, b)$
- Few-shot examples helped the AI identify the correct formula.

question 4: Zero-Shot Prompting (Binary to Decimal Conversion)

Write a zero-shot prompt to generate a Python function that converts a binary number to decimal.

Task:

Test with valid and invalid binary inputs.

Identify missing validation logic.

Prompt

Write a Python function to convert a binary number to its decimal equivalent.

Code

```
#Write a Python function to convert a binary number to its decimal equivalent.
def binary_to_decimal(binary_str):
    decimal = 0
    power = 0
    for digit in reversed(binary_str):
        if digit == '1':
            decimal += 2 ** power
        power += 1
    return decimal

# Example usage
binary_number = "1010"
decimal_number = binary_to_decimal(binary_number)
print(f"The decimal equivalent of binary {binary_number} is {decimal_number}")
```

Code output

```
PS C:\Users\konda\Downloads\AI Assistant> & C:/Users/konda/AppData/Local/
Microsoft/OneDrive/Updater/Updater.exe /start/day3.py"
The decimal equivalent of binary 1010 is 10
```

Explanation

- Uses Python's built-in `int()` with base 2.
- Converts binary string directly to decimal.

question 5: One-Shot Prompting (Decimal to Binary Conversion)

Write a one-shot prompt with an example to generate a Python function that converts a decimal number to binary.

Example:

Input: 10 → Output: 1010

Task:

Compare clarity with zero-shot output.

Analyze handling of zero and negative numbers.

Prompt

Write a Python function to convert a decimal number to binary.

Code

```
day3.py / ...
#Write a Python function to convert a decimal number to binary.
def decimal_to_binary(decimal_num):
    if decimal_num == 0:
        return "0"
    binary = ""
    while decimal_num > 0:
        binary = str(decimal_num % 2) + binary
        decimal_num //= 2
    return binary

(variable) decimal_number: Literal[10]
decimal_number = 10
binary_number = decimal_to_binary(decimal_number)
print(f"The binary equivalent of decimal {decimal_number} is {binary_number}")
```

Code output

```
PS C:\Users\konda\Downloads\AI Assistant> & C:/Users/konda/AppData/Local/Programs/Python/Python39-6/Python.exe -i
The binary equivalent of decimal 10 is 1010
```

Explanation

- `bin(n)` converts number to binary with prefix `0b`
- `[2:]` removes `0b`

question 6: Few-Shot Prompting (Harshad Number Check)

Write a few-shot prompt to generate a Python function that checks whether a number is a Harshad (Niven) number.

Examples:

Input: 18 → Output: Harshad Number

Input: 21 → Output: Harshad Number

Input: 19 → Output: Not a Harshad Number

Task:

Test boundary conditions.

Evaluate robustness

Prompt

Write a Python function to check whether a number is a Harshad number.

Code

```
#Write a Python function to check whether a number is a Harshad number.
def is_harshad_number(n):
    if n == 0:
        return False
    digit_sum = sum(int(digit) for digit in str(n))
    return n % digit_sum == 0

# Example usage
number = 18
if is_harshad_number(number):
    print(f"{number} is a Harshad number.")
else:
    print(f"{number} is not a Harshad number.")
```

Code output

```
PS C:\Users\konda\Downloads\AI ASSISTANT> & C:/Users/konda/AppData/Local/Microsoft/WindowsApps/python3.9/python.exe day3.py
18 is a Harshad number.
```

Explanation

- Calculates sum of digits
- Checks divisibility of number by digit sum
- If divisible → Harshad Number